CS714 Mid-semester test – Practice questions

Question 1: Data pre-processing [5 marks]

Describe 2 approaches to deal with missing data in a tabular dataset. For each approach, give an example of a situation in which the approach would be preferred, and why.

Question 2: Deep Neural Networks and generalisation [5 marks]

For the following situations, describe a simple solution and the intuition behind it:

i. Your model has a high bias after training. (2.5 marks)

ii. Your model has a low bias and a high variance after training. (2.5 marks)

Question 3: Design choices [6 marks]

You trained a simple neural network with the following settings: 2 hidden layers with ReLU units, 5 neurons per layer, random initialisation, trained for 2000 epochs, with L1 regularisation for each hidden layer. Unfortunately, both your training and validation accuracies are equal to 50%, while you know other models performed up to 95% accuracy on the same problem. For each of the following suggestions, explain if it is likely to improve the validation accuracy and why, if implemented in the next iteration of the training.

- Increasing the size of the neural network, i.e., adding more layers and/or more neurons per layer.
- Initialising the weights to 0.
- Using dropout in each layer.

Question 4: Evaluation [3 marks]

Consider the following confusion matrix, resulting from the evaluation of a model on a test set containing 100 instances.

		True class	
		Positive	Negative
Predicted class	Positive	5	20
	Negative	10	65

Calculate the accuracy and recall. What do you think about this model performance?

Question 4: Batch normalisation [5 marks]

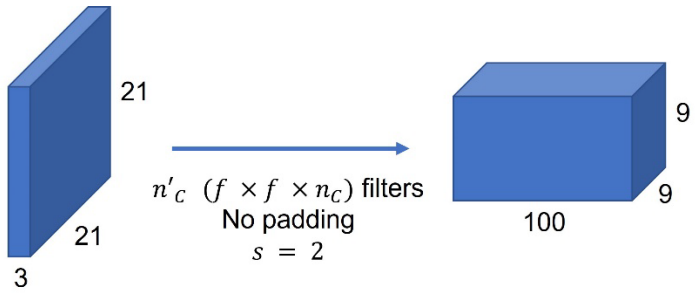
Explain 2 of the effects batch normalisation has on learning in a Neural Network.

Question 5: Attention mechanism and Transformers [4 marks]

a) Explain in your own words what is multi-head attention. (2 marks)
b) Explain why multi-head attention is useful for Transformer models. (2 marks)

Question 6: Convolutional Neural Networks [5 marks]

a) Given the following representation of a convolutional layer with input and output feature maps:



What are the values of the following hyperparameters (explain your workings):

i. The number of filters n'_C (0.5 mark)
ii. The filter size f (1 mark)
iii. The filter width n_C (0.5 mark)

b) Consider a Convolutional Neural Network trained to classify objects in an image. After training, if you were to visualise the neurons' functions at different depths in the network, which neurons are more likely to behave like edge detectors? Explain your reasoning. (3 marks)